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# The long environmental shadow of war

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**Fuel and water infrastructure are increasingly becoming the targets of military conflicts. These attacks pose not only a direct threat to civilian health and safety, but an ongoing threat to the environment that may persist for decades. Military strikes on civilian-critical infrastructure must be condemned if we are to build a just and sustainable future.**

**Explosions following strikes on the Tehran Oil Refinery, 7 March 2026.**

CREDIT: ATTA KENARE/GETTY

On 7 March 2026, US and Israeli airstrikes targeted multiple oil depots and refineries surrounding Iran's capital, Tehran. The result was days of heavy black smoke obscuring the sky and oil-laden acid rain falling across the city of more than 10 million inhabitants. The fallout for the health of Tehran's residents was immediate: the smoke that blanketed the city was highly toxic, including sulfur oxides, nitrogen oxides, polycyclic aromatic hydrocarbons and benzene. Inhalation of sufficiently small particulates (for example, $PM_{2.5}$ or smaller) is a well-documented health hazard, commonly reported from wildfire smoke, that can cause cardiovascular diseases, impact cognitive function and even induce strokes[1]. The black rain that fell over the next days posed a short-term health threat to humans, livestock and wildlife, but may also have a long-term impact on the regional environment and the communities who rely on it. This black rain distils toxins released into the atmosphere from the fuel depot and refinery explosions and subsequent fires and distributes these chemicals across the landscape. This can result in the broad contamination of agricultural soil, surface water and potentially groundwater supplies, as well as fragile ecosystems. Many of the toxins can persist over long time periods in the environment and may be transferred into agricultural products and drinking water for years or decades to come.

This is part of a pattern repeating across conflict zones, in which critical infrastructure is deliberately targeted at great cost to society, human health and the environment. Fuel depots, oil refineries and gas fields have been targeted both by US and Israeli forces in Iran, as well as by Iranian government forces across neighbouring Gulf states. From the burning of the Kuwaiti oil fields in 1991 to recent Ukrainian drone attacks on a Russian oil refinery in Tuapse, the ecological consequences are substantial. Oil-laden black rain from the Tuapse attacks has been reported to have had a direct impact on domestic animals and wildlife, while reports suggest petroleum is leaking from the damaged refinery directly into the Tuapse River, which drains into the Black Sea, where it could further disrupt the marine environment.

Pollution from attacks on oil and gas infrastructure can result in indirect impacts on local and regional water supplies. However, in recent conflicts we have seen a direct impact on critical water infrastructure. In April, Médecins Sans Frontières released a report on the weaponization of water access in Israel's ongoing war in Gaza through the deliberate destruction of and damage to water infrastructure in the Gaza Strip[2]. The World Bank, European Union and United Nations report that nearly 90% of Gaza's water infrastructure has been damaged, including desalination plants, borehole wells, pipelines and sewage systems[3]. In addition to water shortages, water infrastructure damage leads to unsafe and unsanitary conditions when heavy rainwaters cannot be adequately managed, resulting in municipal waste in flood waters and polluted water supplies and local ecosystems. Desalination plants have also been damaged across the Gulf states and Iran since the 2026 conflict broke out, with allegations of deliberate targeting. Desalination plants contribute a significant portion of the water supply in the region, up to 90% in the case of Kuwait, and the debilitation of any of these plants would have immediate and significant human impact. The current conflicts in West Asia are occurring in a region highly vulnerable to changing climate and, in the case of Iran, already experiencing an acute water crisis. The pathways out of this crisis, as Shokri et al. argue in a Comment in this issue of *Nature Sustainability*, depend on proactive investment and cooperation to enhance water resilience and security, maintenance of water infrastructure and smart water governance. These pathways are fundamentally underpinned by a stable and just economic and international order.

The impacts of war on critical infrastructure and the reverberations in the environment are not new; in 2023, *Nature Sustainability* published an editorial on Russian military action in Ukraine and on Ukraine's subsequent ability to safely manage its water resources[4]. As military strikes on basic energy and water infrastructure mount and become increasingly normalized, the distinction between the 'environmental' and 'civilian' casualties of war collapses. Poisoned aquifers, contaminated soils and destroyed desalination plants are attacks on civilians, the impacts of which will be felt for decades to come. In the face of such ongoing destruction, the international community, and sustainability scientists and practitioners in particular, cannot and should not remain silent. A recent letter to *Science* calls on the scientific community to begin tracking and reporting on the environmental impact of war[5]. This is an important first step, but more drastic action is urgently needed. Nations engaged in war must stop direct and indirect strikes on critical water and energy infrastructure. Failing that, international organizations committed to environmental stewardship and human health and nations pledged to the United Nations Sustainable Development Goals must take meaningful steps to condemn and end such actions if we are to build a just and sustainable future for our planet.

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